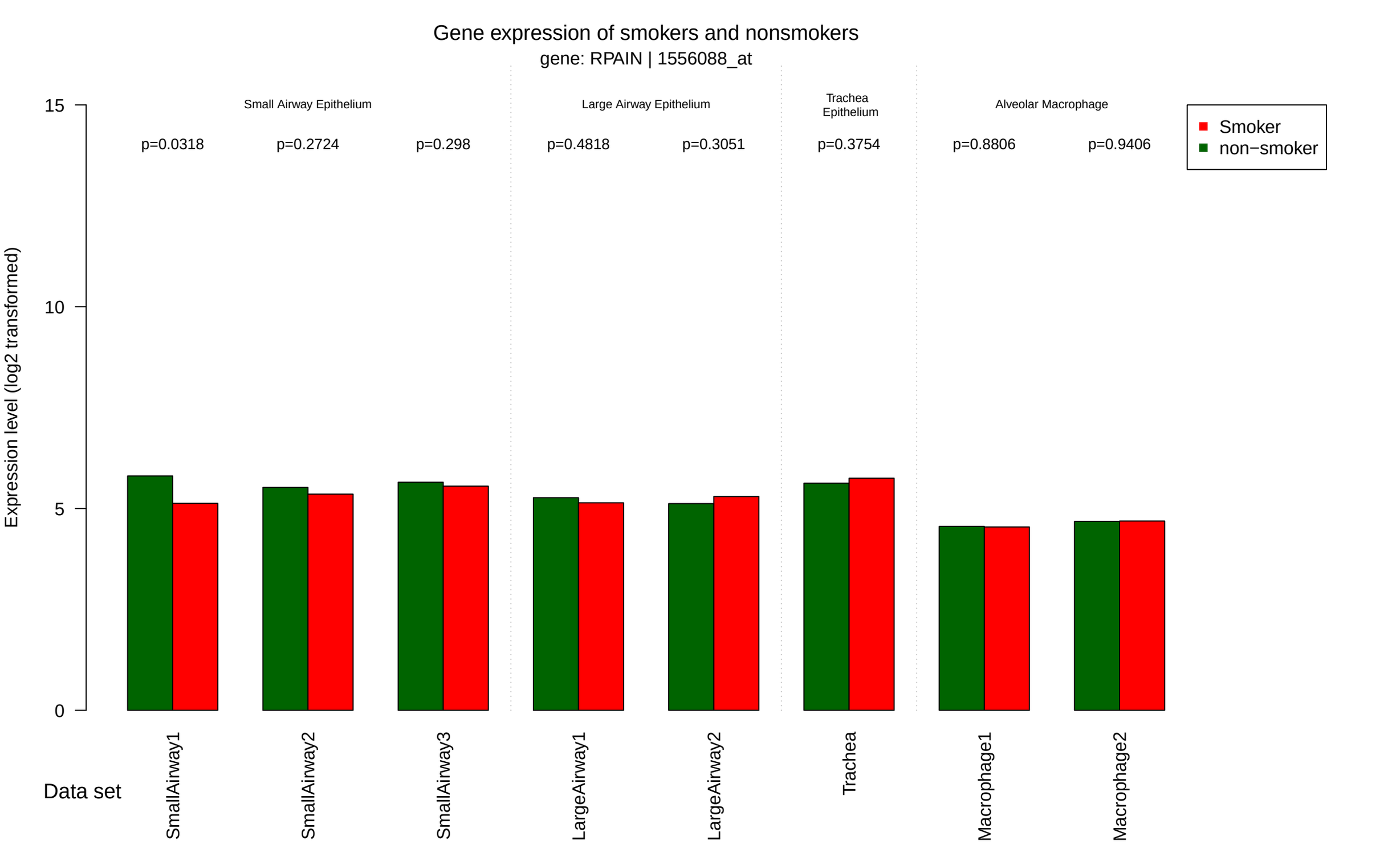
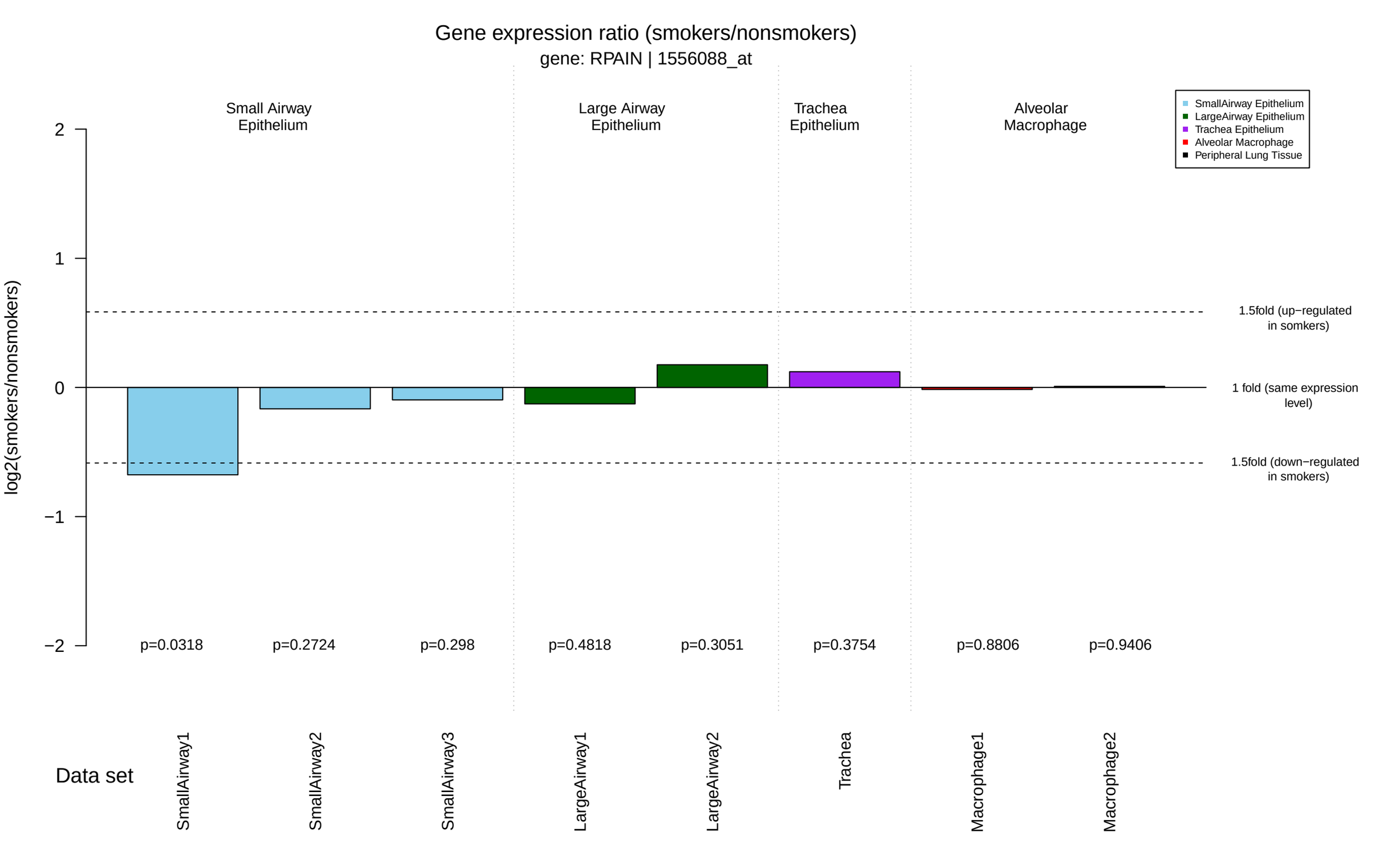
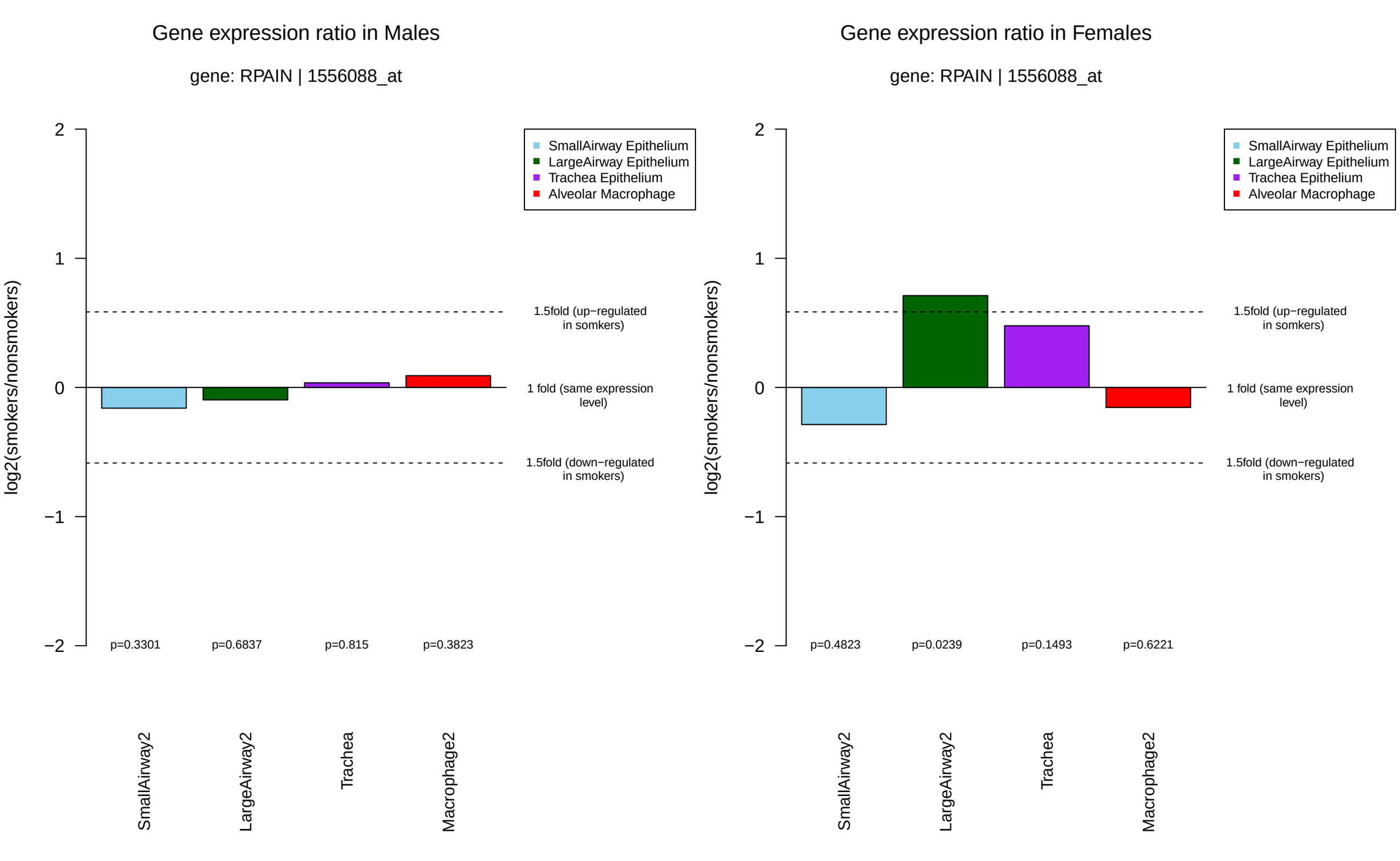
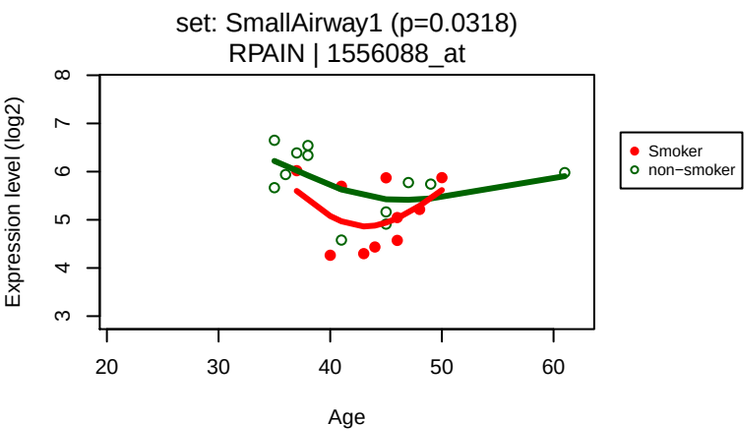
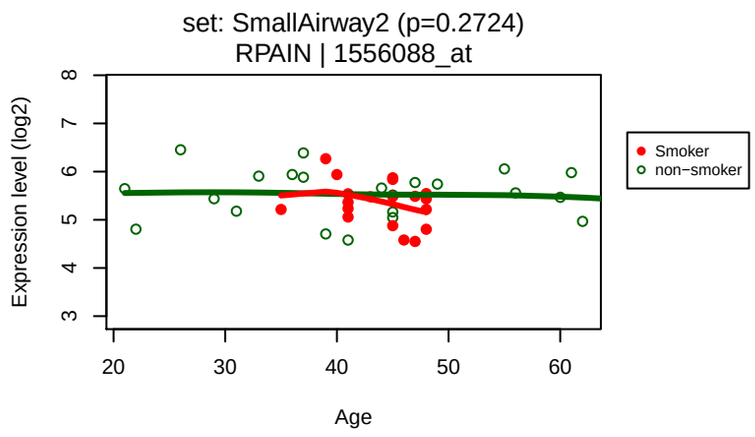
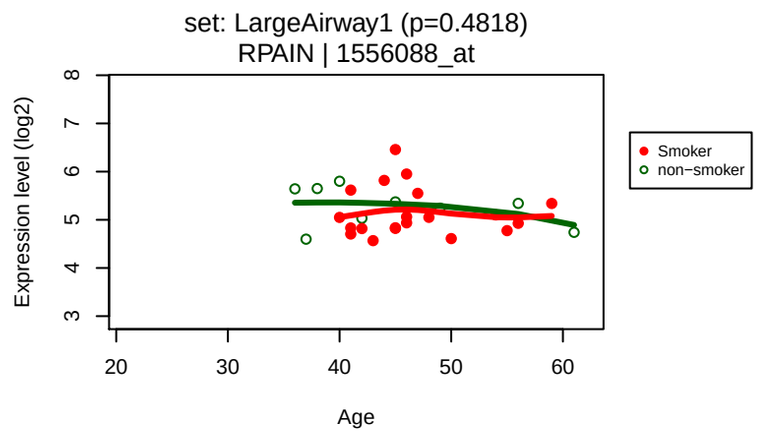

Gene expression of smokers and nonsmokers
gene: RPAIN | 1556088_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0318
p=0.2724
p=0.298
p=0.4818
p=0.3051
p=0.3754
p=0.8806
p=0.9406
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

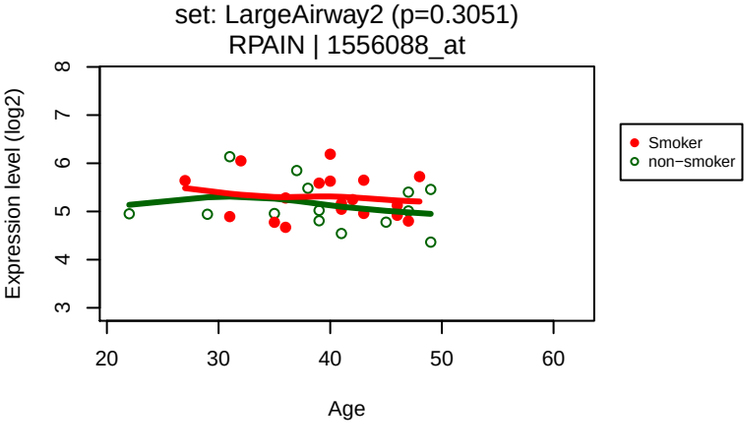

Gene expression ratio (smokers/nonsmokers)
gene: RPAIN | 1556088_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0318
p=0.2724
p=0.298
p=0.4818
p=0.3051
p=0.3754
p=0.8806
p=0.9406
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

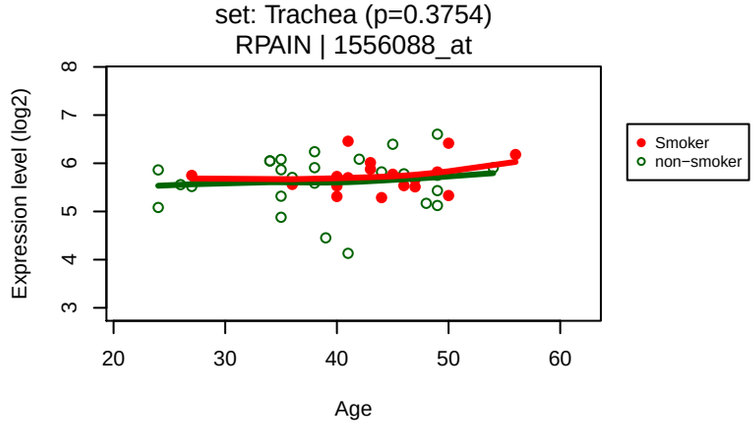

Gene expression ratio in Males
gene: RPAIN | 1556088_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3301
p=0.6837
p=0.815
p=0.3823
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPAIN | 1556088_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4823
p=0.0239
p=0.1493
p=0.6221
SmallAirway2
LargeAirway2
Trachea
Macrophage2

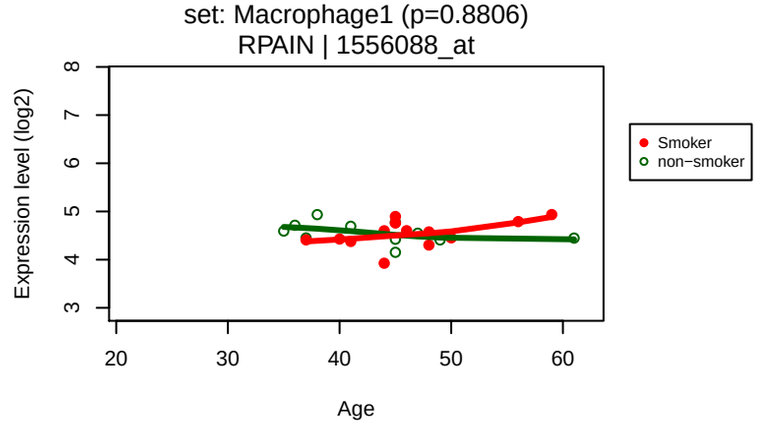

set: SmallAirway1 (p=0.0318)
RPAIN | 1556088_at
Expression level (log2)
Age
Smoker
non−smoker

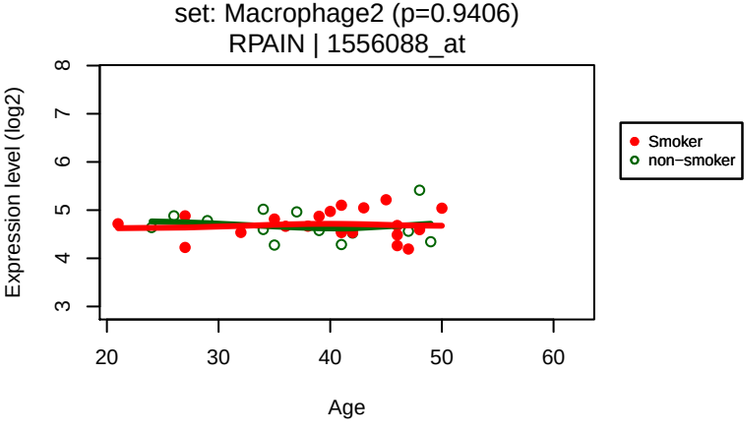

set: SmallAirway2 (p=0.2724)
RPAIN | 1556088_at
Expression level (log2)
Age
Smoker
non−smoker

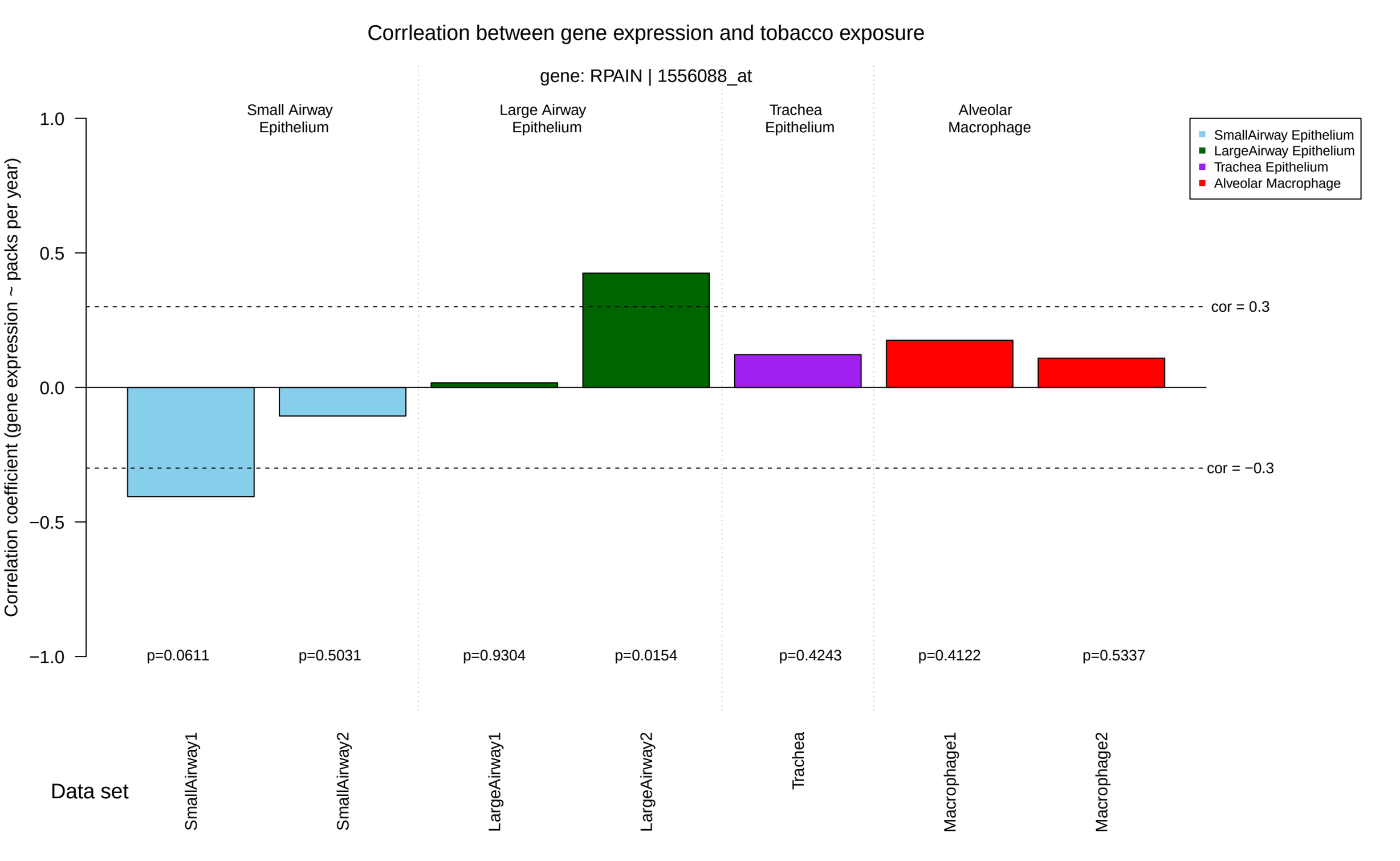

set: LargeAirway1 (p=0.4818)
RPAIN | 1556088_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3051)
RPAIN | 1556088_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3754)
RPAIN | 1556088_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8806)
RPAIN | 1556088_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9406)
RPAIN | 1556088_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPAIN | 1556088_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0611
p=0.5031
p=0.9304
p=0.0154
p=0.4243
p=0.4122
p=0.5337
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2